De novo synthesis of fatty acids

- De novo synthesis of fatty acids occurs in liver, kidney, adipose tissue & lactating mammary gland.
- Enzymes are located in cytosomal fraction of the cell.
- It is called as extramitochondrial or cytoplasmic fatty acid synthase system.

- Major fatty acid synthesized de novo is palmitic acid (16C saturated fatty acid).
- It occurs in liver, adipose tissue, kidney, brain & lactating mammary glands.
- Acetyl CoA is the source of carbon atoms.

- NADPH provides reducing equivalents – NADPH is produced from HMP shunt & malic enzyme reaction.
- Every molecule of acetyl CoA delivered to cytoplasm, one molecule of NADPH is formed.
- ATP supplies energy.

Stages

- Production of acetyl CoA & NADPH
- Conversion of acetyl CoA to malonyl CoA
- Reactions of fatty acid synthase complex.

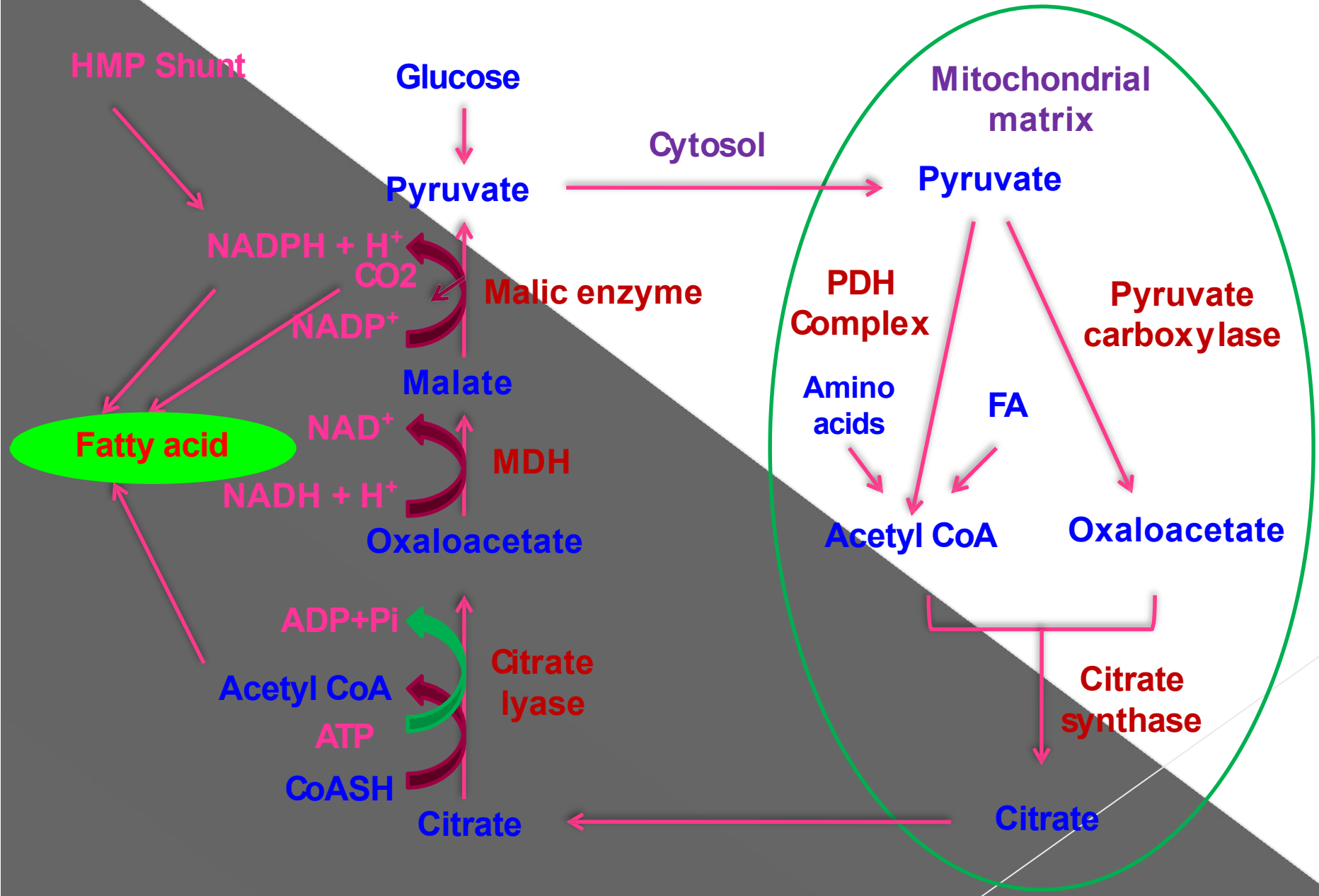
Production of acetyl CoA & NADPH

- Acetyl CoA is the starting material for de novo synthesis of fatty acids.
- Acetyl CoA is produced in the mitochondria by the oxidation of pyruvate, fatty acids, degradation of carbon skeleton of certain amino acids & from ketone bodies.
- Mitochondria are not permeable to acetyl CoA.

- An alternate or a bypass arrangement is made for the transfer of acetyl CoA to cytosol.
- Acetyl CoA condenses with oxaloacetate in mitochondria to form citrate.
- Citrate is freely transported to cytosol by tricarboxylic acid transporter.

- In cytosol it is cleaved by ATP citrate lyase to liberate acetyl CoA & oxaloacetate.
- Oxaloacetate in the cytosol is converted to malate.
- Malic enzyme converts malate to pyruvate.
- NADPH & CO_2 are generated in this reaction.
- Both of them are utilized for fatty acid synthesis

Transfer of acetyl CoA from mitochondria to cytosol

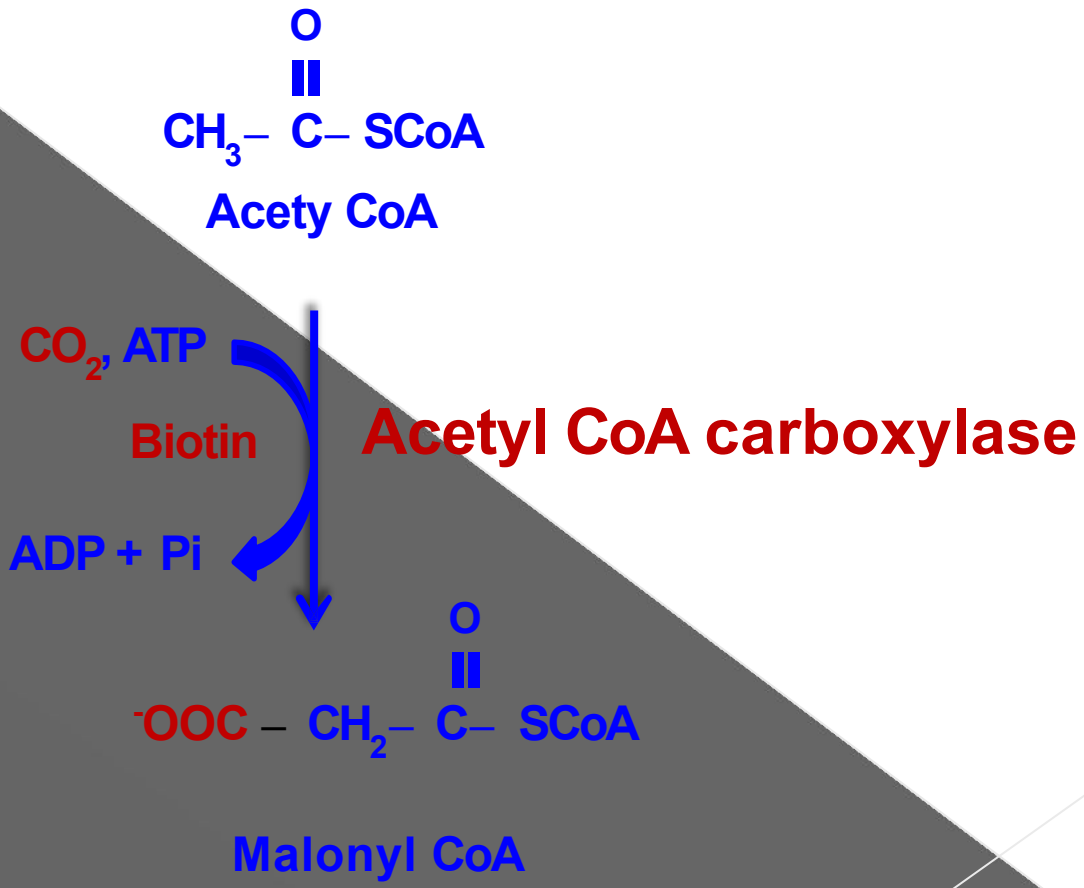


- **Advantages of coupled transport of acetyl CoA & NADPH**
- **The transport of acetyl CoA from mitochondria to cytosol is coupled with the cytosomal production of NADPH & CO_2 which is highly advantageous to the cell for optimum synthesis of fatty acids**

Formation of malonyl CoA

- Acetyl CoA is carboxylated to malonyl CoA by the enzyme acetyl CoA carboxylase.
- This is an ATP-dependent reaction & requires biotin for CO_2 fixation.
- The mechanism of action of acetyl CoA carboxylase is similar to that of pyruvate carboxylase.
- Acetyl CoA carboxylase is a regulatory enzyme

Conversion of acetyl CoA to Malonyl CoA



Reactions of fatty acid synthase complex

- Fatty acid synthase (FAS) - multifunctional enzyme.
- In eukaryotic cells, fatty acid synthase exists as a dimer with two identical units.
- Each monomer possesses the activities of seven different enzymes & an acyl carrier protein (ACP) bound to 4'-phosphopantetheine.
- Fatty acid synthase functions as a single unit catalyzing all the seven reactions.

Advantages of Multi-enzyme complex

- Intermediates of the reaction can easily interact with the active sites of the enzymes.
- One gene codes all the enzymes; all enzymes are in equimolecular concentrations.
- The efficiency of the process is enhanced.

FAS Complex

- First domain or Condensing unit:
- It is initial substrate binding site.
- The enzymes involved are β-keto acyl synthase or condensing enzyme (CE), acetyl transferase (AT) & malonyl transacylase (MT).

Second domain or Reduction unit

- It contains the dehydratase (DH), enoyl reductase (ER), β-keto acyl reductase (KR) & acyl carrier protein (ACP)
- The acyl carrier protein is a polypeptide chain having a phospho-pantotheine group, to which acyl groups are attached in thioester linkage.
- ACP acts like CoA carrying fatty acyl groups.

Third domain or releasing unit

- It is involved in the release of synthesized fatty acid in the cytosol.
- Major fatty acid synthesized is palmitic acid.
- It contains thio-esterase(TE) or de-acylase.

Reactions

- The two carbon fragment of acetyl CoA is transferred to ACP of fatty acid synthase, catalyzed by the enzyme - acetyl CoA-ACP transacylase.
- The acetyl unit is then transferred from ACP to cysteine residue of the enzyme.
- The ACP site falls vacant.

- The enzyme malonyl CoA-ACP transacylase transfers malonate from malonyl CoA to bind to ACP.
- The acetyl unit attached to cysteine is transferred to malonyl group (bound to ACP).
- The malonyl moiety loses CO_2 which was added by acetyl CoA carboxylase.
- CO_2 is never incorporated into fatty acid carbon chain.

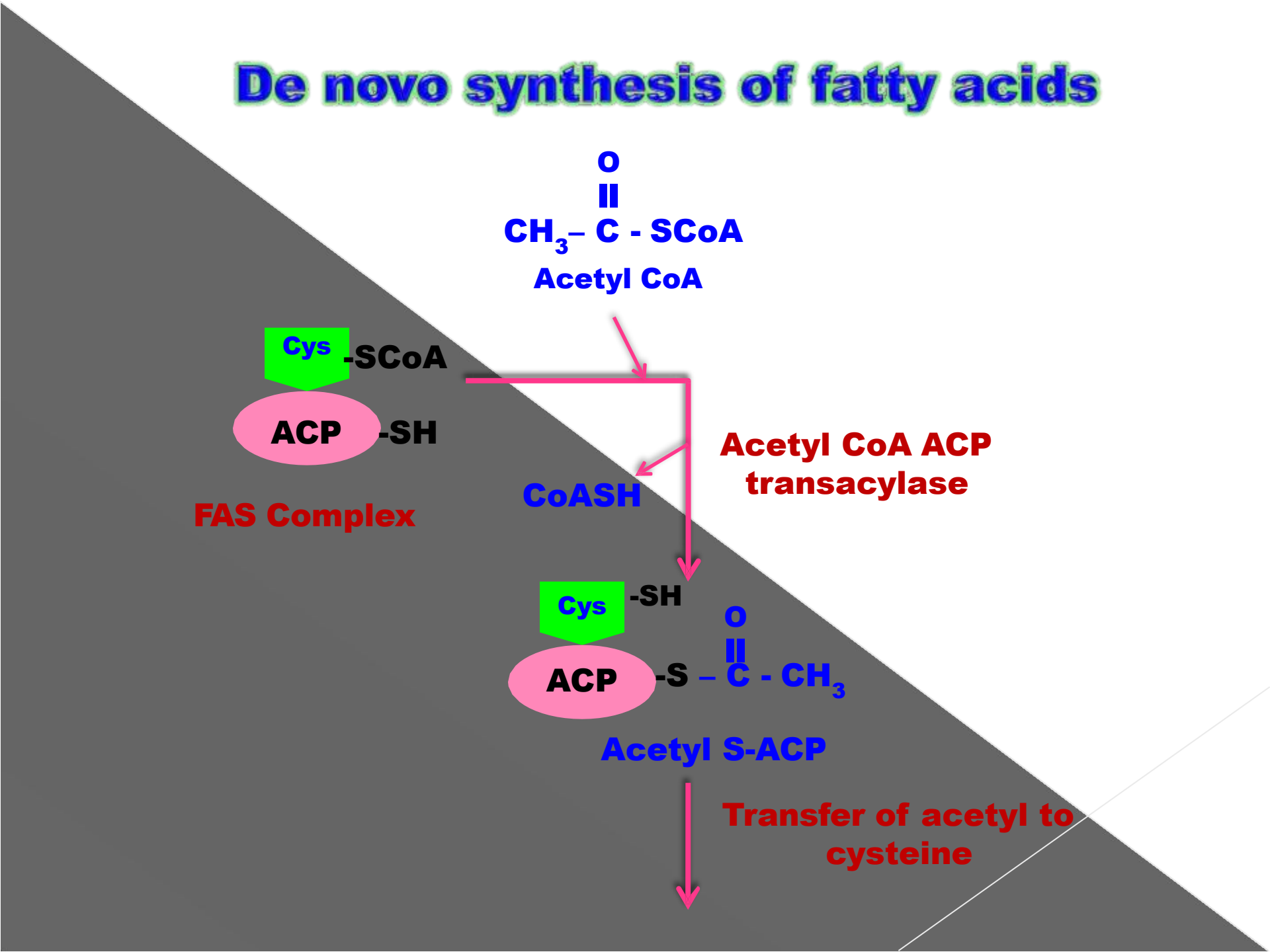
- The decarboxylation is accompanied by loss of free energy which allows the reaction to proceed forward.
- It is catalyzed by β-ketoacyl ACP synthase.
- β -Ketoacyl ACP reductase reduces ketoacyl group to hydroxyacyl group.
- The reducing equivalents are supplied by NADPH.
- β -Hydroxyacyl ACP undergoes dehydration.

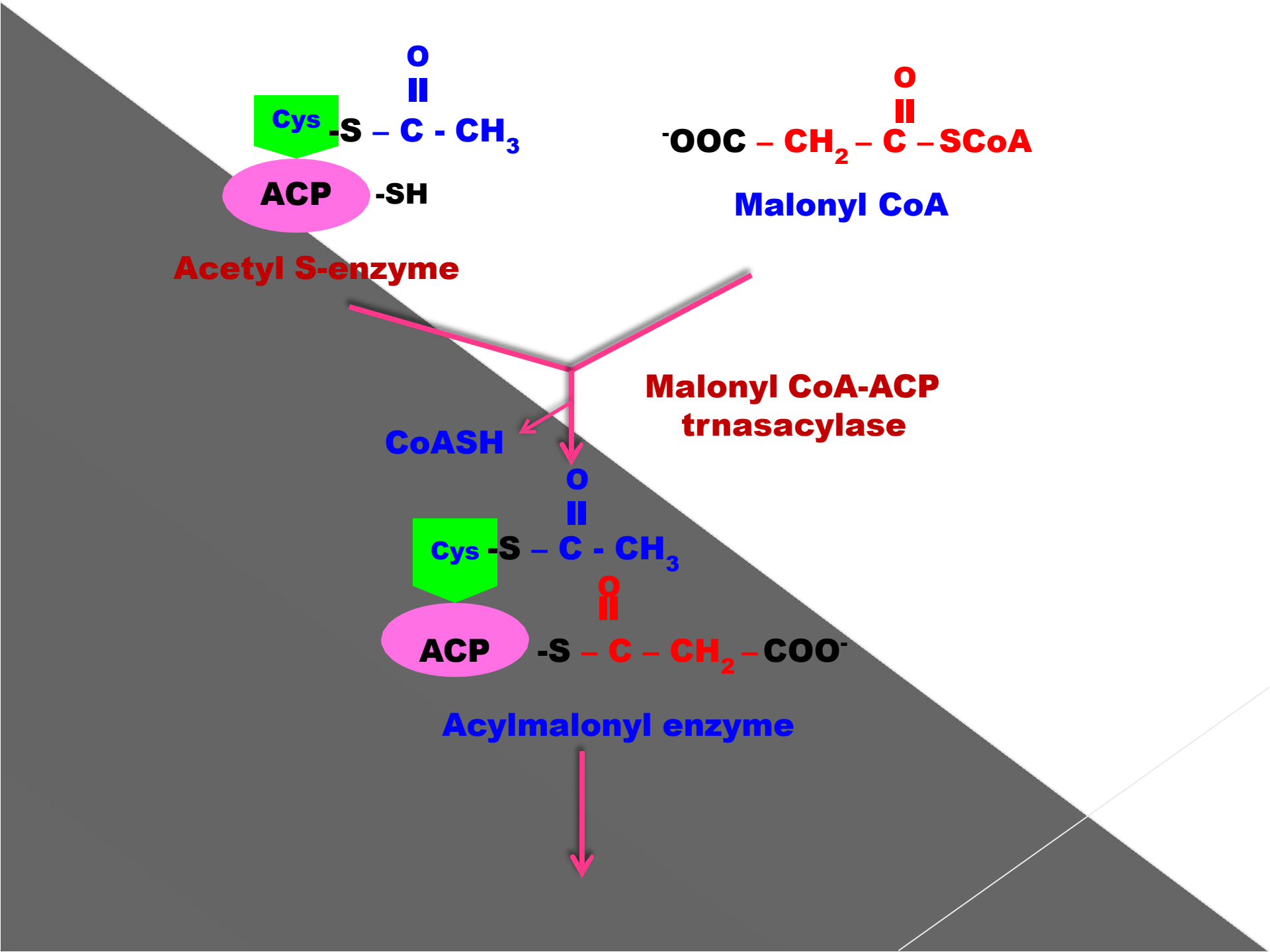
- A molecule of water is eliminated & a double bond is introduced between α & β carbons.
- A second NADPH-dependent reduction, catalysed by enoyl-ACP reductase occurs to produce acyl-ACP.
- The four-carbon unit attached to ACP is butyryl group.

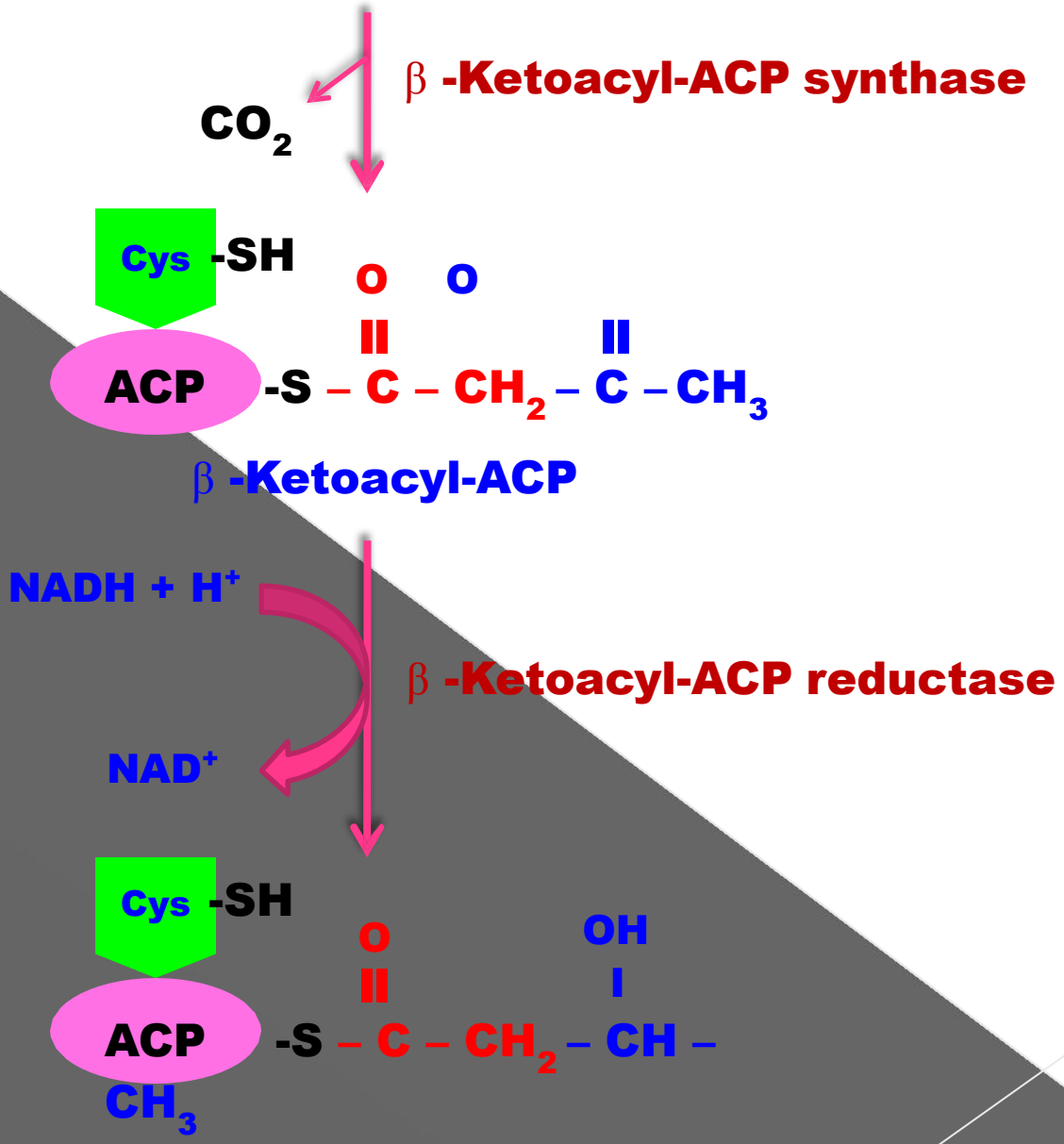
- The carbon chain attached to ACP is transferred to cysteine residue & the reactions of malonyl CoA-ACP transacylase & enoyl-ACP reductase are repeated 6more times.
- Each time, the fatty acid chain is lengthened by a two-carbon unit (obtained from malonyl CoA).

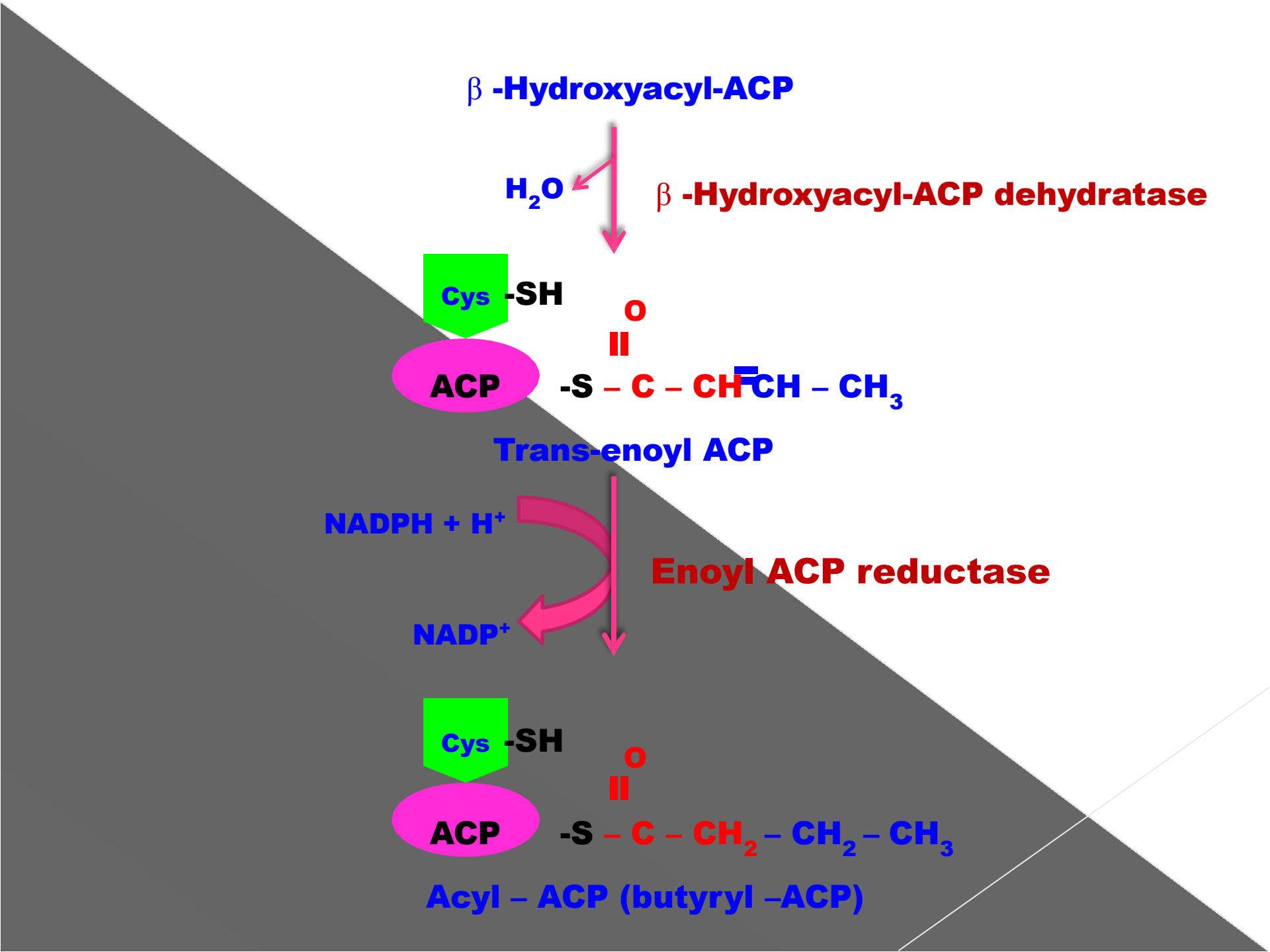
- At the end of 7cycles, the fatty acid synthesis is complete & a 16-carbon fully saturated fatty acid-namely palmitate-bound to ACP is produced.
- The enzyme palmitoyl thioesterase separates palmitate from fatty acid synthase.
- This completes the synthesis of palmitate

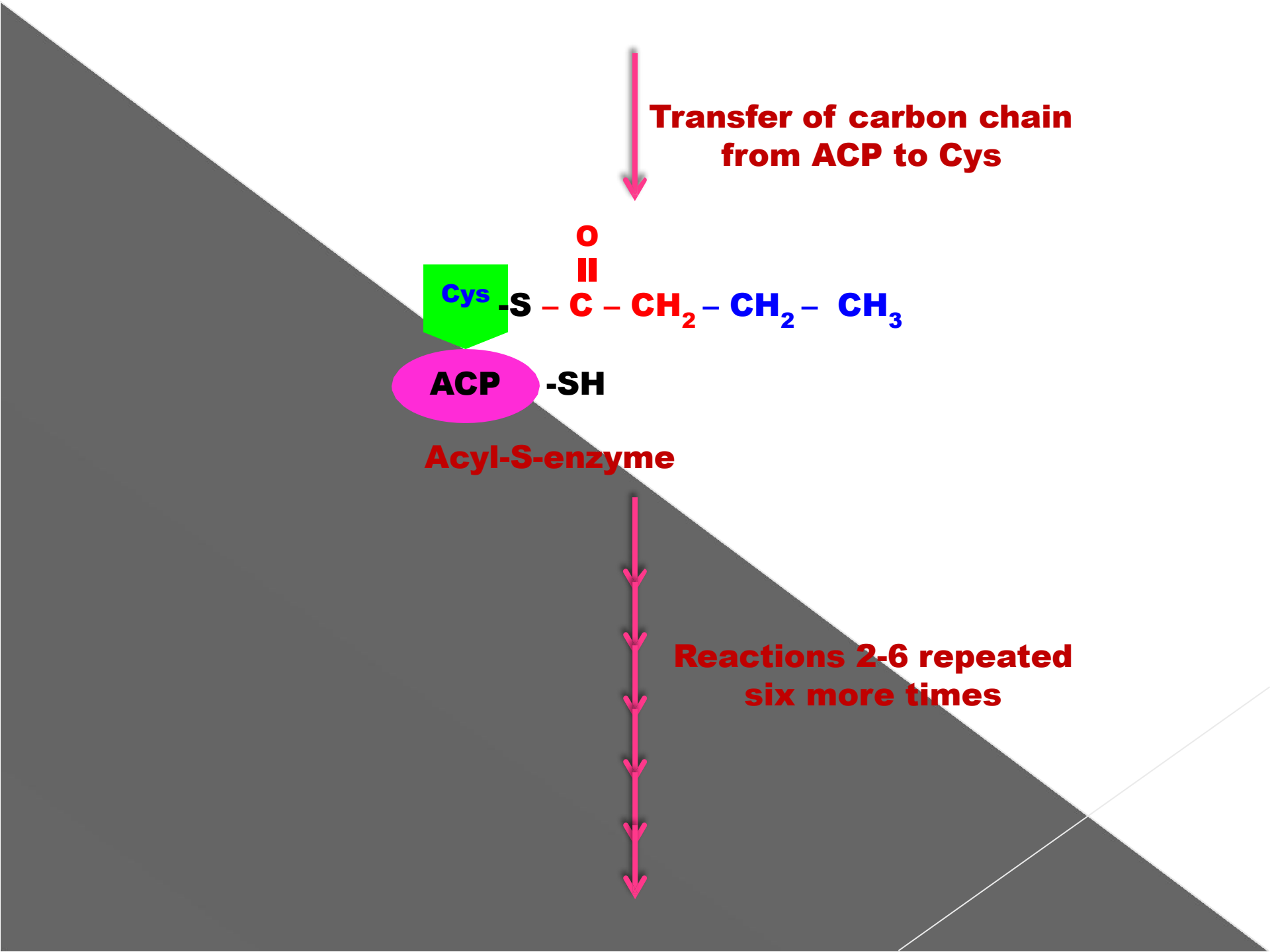
De novo synthesis of fatty acids

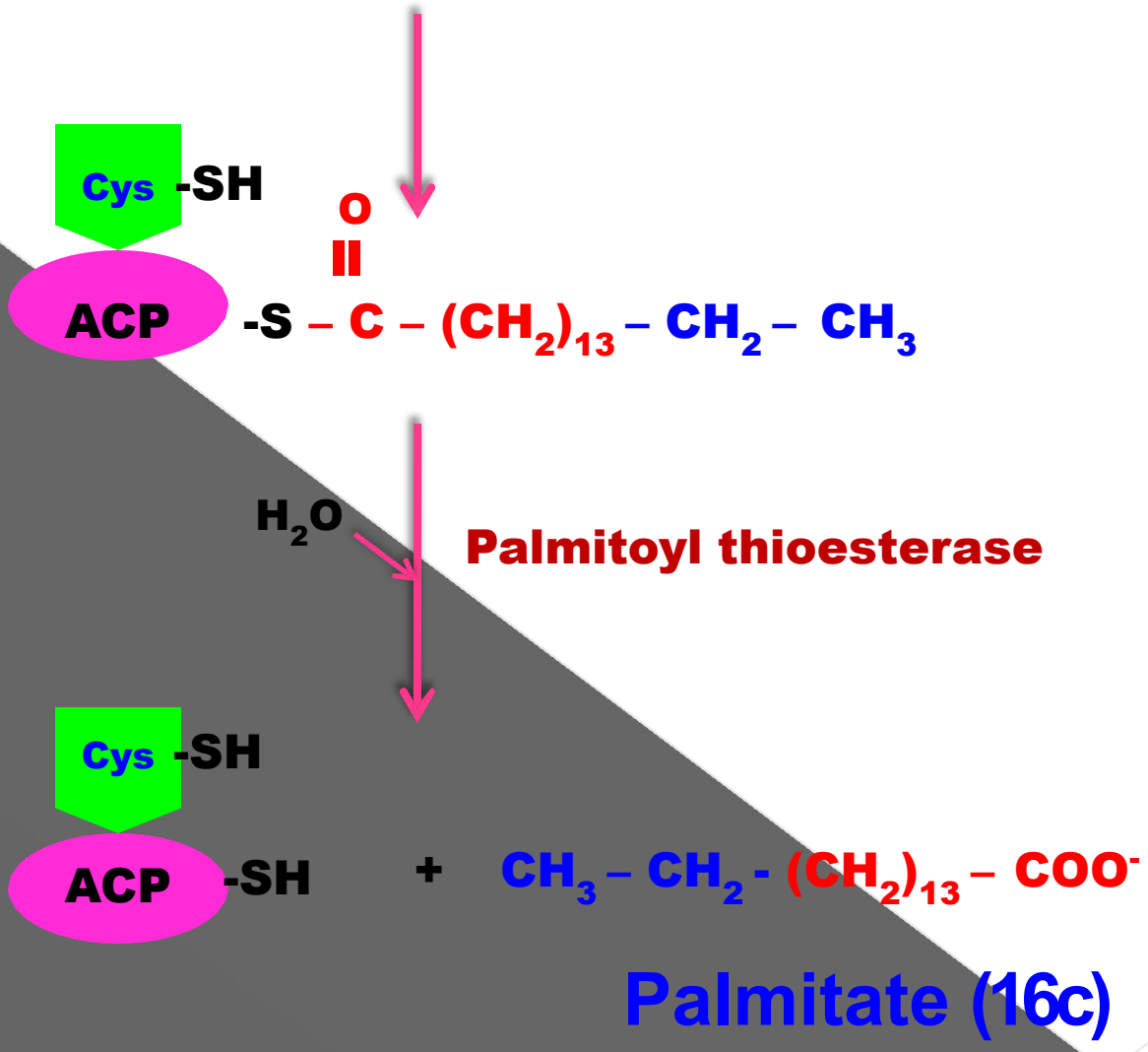



Cys -S – C - CH3
ACP -SH
Acetyl S-enzyme
-OOC – CH2 – C – SCoA
Malonyl CoA
Malonyl CoA-ACP
trnasacylase
CoASH
Cys -S – C - CH3
ACP -S – C – CH2 – COO-
Acylmalonyl enzyme


β -Ketoacyl-ACP synthase
CO2
Cys -SH
ACP
-S – C – CH2 – C – CH3
β -Ketoacyl-ACP
NADH + H+
β -Ketoacyl-ACP reductase
NAD+
Cys -SH
ACP
-S – C – CH2 – CH –
CH3
β -Hydroxyacyl-ACP


β -Hydroxyacyl-ACP
H2O
β -Hydroxyacyl-ACP dehydratase
Cys -SH
ACP
-S – C – CH=CH – CH3
O
Trans-enoyl ACP
NADPH + H+
NADP+
Enoyl ACP reductase
Cys -SH
ACP
-S – C – CH2 – CH2 – CH3
O
Acyl – ACP (butyryl –ACP)

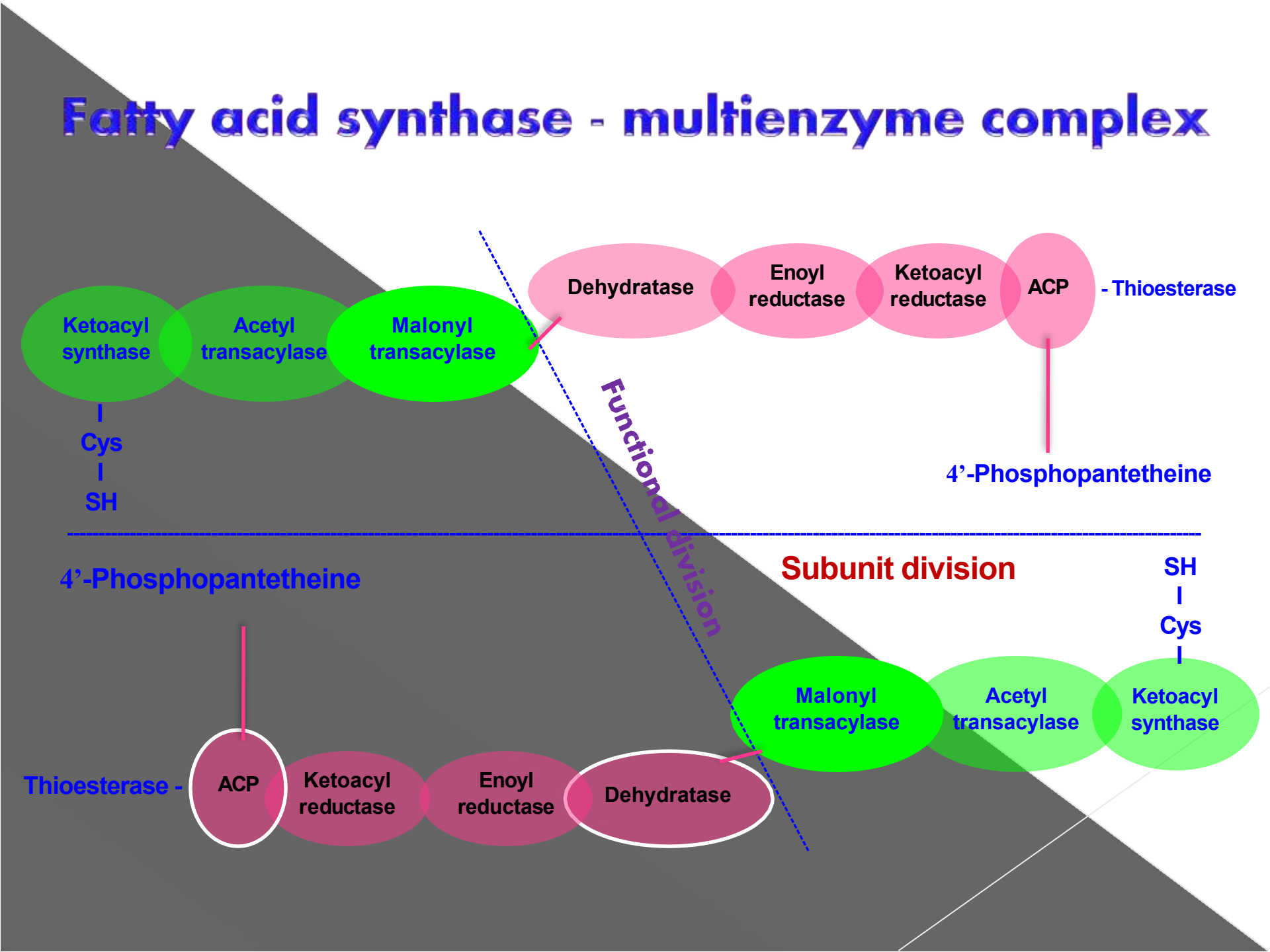

Transfer of carbon chain
from ACP to Cys
O
Cys -S – C – CH2 – CH2 – CH3
ACP -SH
Acyl-S-enzyme
Reactions 2-6 repeated
six more times


Cys -SH
ACP
O
-S – C – (CH2)13 – CH2 – CH3
H2O
Palmitoyl thioesterase
Cys -SH
ACP -SH
+ CH3 – CH2 - (CH2)13 – COO-
Palmitate (16c)

Fatty acid synthase complex

- It is a multienzyme complex
- Fatty acid synthase is a dimer composed of two identical subunits (monomers),
- Each with a molecular weight of 240,000.
- Each subunit contains the activities of 7 enzymes of FAS & an ACP with 4'-phosphopantetheine SH group.
- The two subunits lie in antiparallel (head to tail) orientation

Fatty acid synthase - multienzyme complex



- The -SH group of phosphopantetheine of one subunit is in close proximity to the -SH of cysteine residue (of the enzyme ketoacyl synthase) of the other subunit.
- Each monomer of FAS contains all the enzyme activities of fatty acid synthesis.
- Only the dimer is functionally active.

- The functional unit consists of half of each subunit interacting with the complementary half of the other.
- FAS structure has both functional division & subunit division
- The two functional subunits of FAS independently operate & synthesize two fatty acids simultaneously

Significance of FAS complex

- The FAS complex offers great efficiency that is free from interference of other cellular reactions for the synthesis of fatty acids.
- There is a good coordination in the synthesis of all enzymes of the FAS complex.

Regulation of fatty acid synthesis

- Fatty acid production is controlled by enzymes, metabolites, end products, hormones and dietary manipulations.
- Acetyl CoA carboxylase:
- This enzyme controls a committed step in fatty acid synthesis.

- Acetyl CoA carboxylase exists as an inactive protomer (monomer) or an active polymer.
- Citrate promotes polymer formation & increases fatty acid synthesis.
- Palmitoyl CoA & malonyl CoA cause depolymerization of the enzyme, inhibits the fatty acid synthesis.

Hormonal influence

- Hormones regulate acetyl CoA carboxylase by a separate mechanism-phosphorylation (inactive form) & dephosphorylation (active form) of the enzyme.
- Glucagon, epinephrine & norepinephrine inactivate the enzyme by cAMP dependent phosphorylation.

- Insulin, dephosphorylates & activates the enzyme.
- Insulin promotes fatty acid synthesis while glucagon inhibits.
- Insulin stimulates tissue uptake of glucose & conversion of pyruvate to acetyl CoA.
- This also facilitates fatty acid formation.

Dietary regulation

- Consumption of high carbohydrate or fat-free diet increases the synthesis of acetyl CoA carboxylase & fatty acid synthase, which promote fatty acid formation.
- Fasting or high fat diet decreases fatty acid production by reducing the synthesis of acetyl CoA carboxylase & FAS.

Availability of NADPH

- The reducing equivalents for fatty acid synthesis are provided by NADPH which come either from citrate (acetyl CoA) transport or hexose monophosphate shunt.
- About 50-60% of required NADPH is obtained from HMP shunt, which significantly influences fatty acid synthesis

Synthesis of long chain fatty acids from Palmitate

- Chain elongation can take place either in mitochondria or in endoplasmic reticulum (microsomes), by separate mechanism.
- The microsomal chain elongation is more predominant and involves successive additions of malonyl CoA with the participation of NADPH.

- A specific group of enzymes - elongases bring about fatty acid chain elongation.
- The mitochondrial chain elongation is almost a reversal of β-oxidation of fatty acids.
- Acetyl CoA molecules are successively added to fatty acid to lengthen the chain.
- The reducing equivalents are derived from NADPH.

References

- Lippincott's Illustrated Review: Biochemistry
- Textbook of Medical Biochemistry: Chatterjea

Thank You